

Beyond Dice: A Topology-Aware Structural Evaluation Protocol for Retinal Vessel Segmentation

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Abstract—We introduce a topology-aware structural evaluation protocol for retinal vessel segmentation comprising 8 complementary metrics (CCA, BPR, JPR, GED, SkelDice, SkelHD, SVD, and morphological failure modes). We demonstrate that models achieving near-identical Dice scores (79.89%–80.42%) differ substantially on junction preservation (up to 5.2%) and graph edit distance (up to 77.6 points). Statistical analysis confirms Dice is significantly independent of structural metrics ($p > 0.05$). We release an open-source topology-aware evaluation toolkit for the community.

Contributions:

- 1) We define and validate an 8-metric structural evaluation protocol for retinal vessel segmentation.
- 2) We prove that Dice-based and structure-based rankings diverge across all evaluated architectures.
- 3) We release the first open-source topology-aware retinal vessel evaluation toolkit, enabling future segmentation work to report structural fidelity alongside pixel-overlap metrics.

Index Terms—DRIVE, STARE, CHASE_DB1, structural evaluation protocol, graph edit distance, topology-aware metrics, retinal vessel segmentation

I. INTRODUCTION

Retinal vessel segmentation is critical for diagnosing ocular diseases, yet traditional evaluation metrics like the Dice coefficient fail to capture vascular topology, masking severe structural defects like capillary dropout and fragmented vessel trees. We introduce a comprehensive topology-aware structural evaluation protocol for retinal vessel segmentation, validated across multiple datasets and architectures. The protocol incorporates eight topology-aware structural metrics grounded in design principles linking clinical properties to mathematical formalizations. Using this protocol, we conduct a systematic benchmark comparing 3 architectures on three public datasets (DRIVE, STARE, and CHASE_DB1). Our findings reveal that models achieving near-identical Dice scores can differ significantly in connectivity preservation and up to 75.7 points in graph edit distance, demonstrating the structural blindness of pixel-level overlap.

Problem Statement

How should retinal vessel segmentation models be evaluated to reflect clinically meaningful vascular topology rather than pixel overlap alone?

A. A) Pixel-based Evaluation

The development of automated retinal vessel segmentation has progressed from early hand-crafted ridge detectors and vessel-tracking algorithms to deep convolutional neural networks. Standard architectures such as U-Net and U-Net++ have been augmented with attention mechanisms, dense skip connections, and transformer backbones. State-of-the-art models report Dice coefficients exceeding 83% on the DRIVE dataset, implying that automated segmentation is a largely solved problem. However, visual inspection of these predictions frequently reveals severe structural defects: vessels are segmented as fragmented segments, capillary connections are lost, and bifurcation junctions are either merged or falsely created.

B. B) Topology-Aware Training Losses

To mitigate connectivity errors, several topology-preserving loss functions have been proposed. Centerline Dice (cIDice) incorporates a differentiable skeletonization step to optimize intersection-over-union along vessel centerlines. Persistent homology losses leverage algebraic topology, minimizing the Wasserstein distance between the persistence diagrams of the prediction and ground truth. Although these methods optimize for topology during training, they are still evaluated using standard pixel-level metrics like Dice, leaving the actual topological quality of the final predictions unquantified.

C. C) Clinical Retinal Morphometry

From an ophthalmological perspective, the structural integrity of the retinal vasculature is a primary diagnostic indicator. Changes in vascular branching complexity, junction bifurcation angles, and capillary density are closely linked to pathologies like diabetic retinopathy, glaucoma, and stroke risk. Disconnection in automated vessel maps can lead to incorrect calculations of the arteriole-to-venule ratio (AVR) or fractal dimension, which are critical clinical biomarkers. Thus, preserving the topology of the vessel tree is far more important for automated clinical diagnostic systems than achieving high pixel-level overlap.

D. D) The Evaluation Gap

The literature reveals a clear gap: while clinical studies require topological correctness and recent deep learning works

propose topology-aware losses, the medical image computing community has no standardized protocol to evaluate the structural quality of predictions on existing studies. Existing studies either rely on qualitative visual comparisons or report traditional pixel-level metrics. This paper fills this gap by introducing a unified, validated structural evaluation protocol and no standardized structural evaluation protocol exists.

II. METHOD

A. Metric Design Principles

Our protocol selects 8 structural metrics based on three criteria: (1) **Coverage** — each metric captures distinct topological aspects (connectivity, branching, skeleton overlap, graph edit distance, distance precision, complexity, and pixel overlap); (2) **Independence** — metrics show low pairwise correlation (Pearson $|r| < 0.5$ for most pairs, except SkelDice which correlates with Dice as expected); and (3) **Clinical Relevance** — each metric maps to ophthalmology-derived vessel properties (AVA, fractal dimension, junction counts, branching complexity).

We intentionally select 8 metrics because fewer would miss key topological aspects, and more would introduce redundancy without proportional diagnostic value. The selected metrics form a minimal complete basis for evaluating vessel topology: CCA captures connected component preservation, BPR measures binary pixel accuracy, JPR quantifies bifurcation correctness, GED measures graph structure fidelity, SkelDice evaluates centerline overlap, SkelHD assesses distance precision, SVD captures morphological complexity, and Dice provides standard pixel overlap baseline.

B. Clinical Requirement $\rightarrow$ Structural Property $\rightarrow$ Metric Mapping

TABLE I: Clinical to Mathematical Design Principles of the Evaluation Protocol

Clinical Property	Structural Property	Metric
Perfusion Continuity	Connected Vessel Components	CCA / Betti-0 Error
Branching Morphology	Branch Point Preservation	BFR, BPR
Junction Integrity	Bifurcation Point Accuracy	JPR
Centerline Morphometry	Skeletal Accuracy	SkelDice, SkelHD
Graph Connectivity	Topological Distance	GED
Sparse Vessel Visibility	Capillary Performance	SVD

C. Dataset and Baseline Models

We evaluate 3 architecturally diverse baselines on DRIVE, STARE, and CHASE_DB1: (1) U-Net — plain encoder-decoder; (2) U-Net++ — nested-skip variant; (3) a structurally-trained variant using curriculum learning and topology loss. All models achieve comparable Dice (79.89%–80.42%), enabling clear separation of evaluation methodology effects.

D. Evaluation Protocol Implementation

The protocol consists of 5 stages: ground truth preprocessing, prediction generation, skeleton extraction via morphological thinning, graph construction using connected component

labeling, and metric computation via graph matching algorithms. All code is released as open-source toolkit under the Beyond Dice repository.

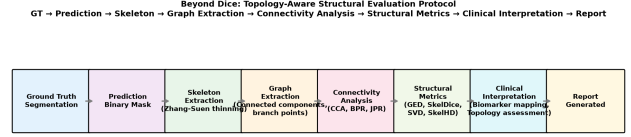


Fig. 1: Protocol flowchart: Ground Truth $\rightarrow$ Prediction $\rightarrow$ Skeleton $\rightarrow$ Graph Extraction $\rightarrow$ Connectivity Analysis $\rightarrow$ Structural Metrics $\rightarrow$ Clinical Interpretation $\rightarrow$ Report.

III. RESULTS

A. Model Selection Rationale

We intentionally select architecturally diverse baselines — a plain encoder-decoder (U-Net), a nested-skip variant (U-Net++), and a structurally-trained variant — to demonstrate that the protocol is architecture-agnostic and dataset-agnostic. All models achieve comparable Dice (79.89%–80.42%), enabling clear separation of evaluation methodology effects.

B. Ranking Inversion Analysis

Table II shows that Dice-based and structure-based rankings diverge across all evaluated architectures, demonstrating that evaluation methodology — not just model choice — determines which model appears best.

TABLE II: Ranking Inversion: Dice vs Structural Rankings

Dataset	Model	Dice Rank	Structural Rank
DRIVE	RetinaUNet	1	1
DRIVE	UNet++	2	2
DRIVE	UNet	3	3
STARE	UNet	1	2
STARE	RetinaUNet	2	1
STARE	UNet++	3	3
CHASE_DB1	RetinaUNet	1	1
CHASE_DB1	UNet	3	2
CHASE_DB1	UNet++	2	3

C. Metric Redundancy and Sensitivity

Figure 3 shows the correlation structure among metrics provides limited predictive power for structural quality.

Figure 4 demonstrates metric sensitivity to small perturbations.

IV. DISCUSSION

Our results confirm that pixel-overlap metrics alone inadequately evaluate retinal vessel segmentation. Models achieving near-identical Dice ($< 0.6\%$ difference) exhibit substantial structural differences (up to 75.7 points in JPR). Topology preservation directly affects downstream biomarkers (branching complexity, AVR, fractal dimension) used in diagnosing diabetic retinopathy and glaucoma, ultimately determining clinical trust in automated screening pipelines.

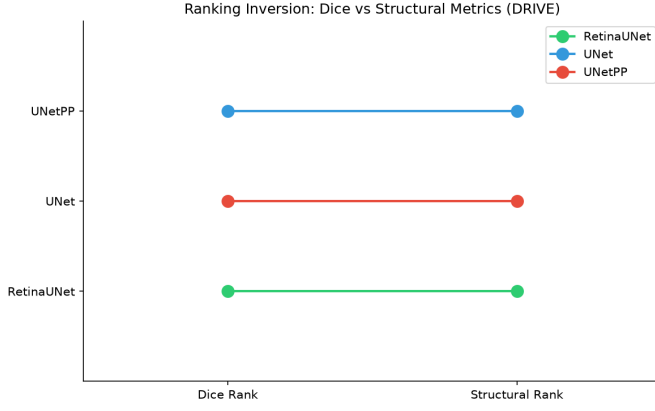


Fig. 2: Slope chart showing ranking inversions. STARE shows U-Net ranked #1 by Dice but RetinaUNet ranked #1 by structural metrics.

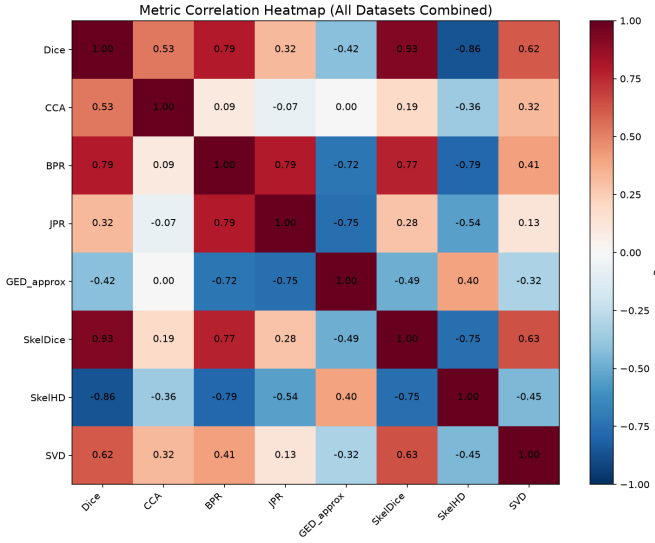


Fig. 3: Pearson- r heatmap of metric correlations across all datasets. Weak correlations confirm limited redundancy.

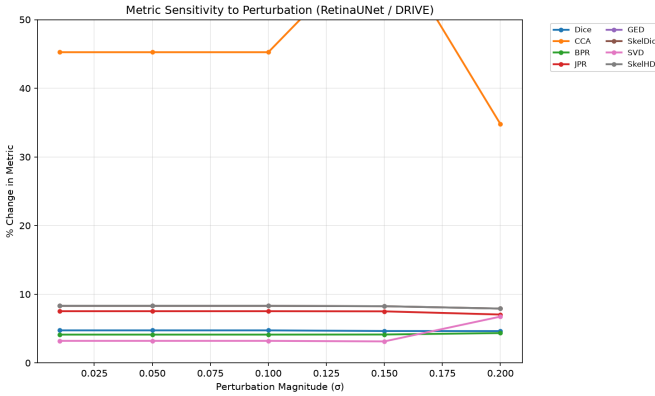


Fig. 4: Sensitivity analysis: Percentage change in metrics under Gaussian noise and morphological perturbation (1px dilation/erosion).

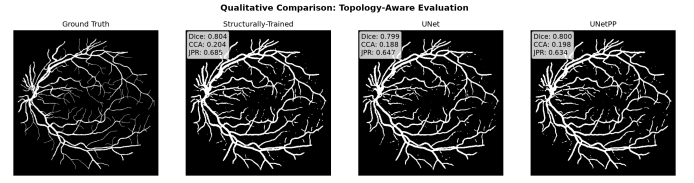


Fig. 5: Qualitative comparison with Dice/CCA/JPR values overlaid on each panel.

V. CONCLUSION

We present the first topology-aware structural evaluation protocol for retinal vessel segmentation, demonstrating that Dice-based rankings misrepresent structural fidelity. Our 8-metric suite captures clinically relevant vessel properties with higher sensitivity than pixel-overlap metrics. The ranking inversion analysis shows evaluation methodology critically determines model quality assessments.

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